MULTI-SENSOR HARDWARE SYNCHRONIZATION & TIMESTAMP METROLOGY

Eliminating cross-modal phase skew between camera photons and IMU dynamics via IEEE 1588 PTP hardware timestamps

❌ FLAWED: NAIVE SOFTWARE TIMESTAMPS (MEASURING APPLICATION ARRIVAL TIME)

Exposure Window	Rolling Shutter Readout	MIPI / PCIe DMA	V4L2 IRQ + Wakeup	t_app = +24.7 ms ⚠️
t_mid = 0.0 ms (Truth)	+16.7 ms (60 Hz)	+4.2 ms (Contention)	+3.8 ms (OS Jitter)	Software Timestamp Taken

Unmeasured Phase Skew Δt = +24.7 ms (Variable Jitter: 12–35 ms)

Consequence for State Estimation: Kalman Filter / Factor Graph fuses camera pose from *t = 0.0 ms* with IMU velocity from *t = +24.7 ms*.
At robot speed *v = 1.5 m/s*, this +24.7 ms timestamp error injects **37.0 mm of fictitious kinematic error**, causing filter divergence and motor instability.

✓ PHYSICAL AI STANDARD: IEEE 1588 PTP HARDWARE EXPOSURE MIDPOINT TIMESTAMPS

PTP GRANDMASTER CLOCK

`IEEE 1588 / PTPv2`

- Microsecond synchronization
- Clock Skew: **< 2 μs**

CAMERA SENSOR (HARDWARE STROBE)

- HW timer latches timestamp at **exposure midpoint**

`t_mid = 1724158920.104821 s (±1 μs)`

- Timestamp stamped into raw metadata packet header

IMU SENSOR (1000 Hz FIFO LATCH)

- High-rate gyroscope & accelerometer ring buffer

`t_imu = 1724158920.104820 s (Exact match)`

- Direct temporal interpolation at exact *t_mid*

Exact Metric Alignment: Even if Linux OS takes 30 ms to deliver the image tensor to Python, the system knows the exact physical photons were captured at *t_mid*.

- Phase Skew reduced from **24.7 ms** to **< 0.005 ms (5 μs)**.
- State estimation error drops from **37.0 mm** to **< 0.01 mm**, enabling millimeter-precision grasping and rock-solid dynamic stability.